

# SANDIP SHAW

## Data Scientist

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## SUMMARY

Data Scientist and ML Engineer with 2+ years of experience delivering production-grade AI systems, analytical pipelines, and data-driven solutions. Specialise in machine learning, predictive analytics, and intelligent automation, with a track record of end-to-end project ownership from data ingestion through model deployment. Currently completing an MSc in Data Science (class topper, graduating August 2026). Proven ability to surface actionable business intelligence, reduce manual effort, and communicate findings to management stakeholders.

## EXPERIENCE

**Data Scientist Intern | AI Developer | Cell Hub MS** | 01/2026 – Present | Pune

*Company specializing in data solutions*

- Engineered an event-driven data provisioning pipeline with 95% mapping accuracy, automatically parsing user-uploaded CSVs into a 50+ table MySQL schema across 9 business domains, standardising data availability across 9 business functions and eliminating manual data entry.
- Delivered a RAG querying pipeline with **90%+** response relevance, serving 20–100 active production users; taken from design through demo cycles to deployment within 1–2 months.
- Collaborated within a 5+ member agile team on sprint planning, code reviews, and technical documentation throughout the product lifecycle.

**Data Analyst | Innover Digital** | 06/2022 – 04/2023 | Kolkata, India

*Digital agency focused on logistics data analytics*

- Analysed truck logistics data across 10+ US-based clients, contributing to an estimated **10–20%** improvement in route efficiency through targeted optimisation opportunities.
- Automated data transformation pipelines using Python and Advanced Excel, saving **12+** hours of recurring reporting effort per week.
- Built **10+** Power BI and Tableau dashboards adopted by US logistics clients; presented findings directly to management stakeholders.

**Data Analyst Intern | Krop Gains** 08/2021 – 03/2022 | Remote

*Remote position analysing agricultural data*

- Scraped and pre-processed **100 GB** of Indian agricultural data from multiple web sources into a uniform schema for downstream analytics.
- Developed a Computer Vision model for automated crop disease classification using a custom CNN trained on 6,000 augmented images across 12 crop categories; implemented data validation and model validation pipelines achieving 79% test accuracy.
- Built 20+ Tableau dashboards covering crop production, cost, and import/export trends; analysis revealed an average **3.65%** performance spike across monitored parameters.

## EDUCATION

**MSc Data Science & Spatial Analytics**

**Symbiosis Institute of Geoinformatics**

06/2024 – 08/2026 | Pune | **Class Topper · CGPA 8.82**

**BSc Data Science**

**Techno India University**

06/2019 – 08/2022 | Kolkata | **CGPA 8.41**

## CERTIFICATIONS

**PW-Skills Full Stack Data Science Pro**

Full Stack Data Science Pro certification

## KEY ACHIEVEMENTS

### Achievements in Data Efficiency

Achieved 95% mapping accuracy in data parsing and contributed to significant efficiency improvements for logistics clients.

## LANGUAGES

**English** Proficient | **Hindi** Native | **Bengali** Native

## SKILLS

**Language & Database:** Python, SQL, MongoDB, MySQL

**Analytics:** Power BI, Tableau, Microsoft Excel, PowerPoint

**Frameworks:** PyTorch, TensorFlow, Karas, Scikit-Learn

**Concepts:** Deep Learning, Machine Learning, Automated Pipeline Development, RAG, NLP, Langchain, Fast API, A/B Testing, Statistical Modelling, Predictive Analytics, Information Retrieval, Data Storytelling, Process Automation

**Soft Skill:** Cross-functional communication, Problem structuring, Collaborative teamwork, Ownership & delivery

## PROJECTS

### AI-Powered Safe Route Recommender

*KDE | ML Flow | Azure | SK Learn | Fast API | Docker | CI/CD*

- Developed a crime-aware route recommender for commuters using FastAPI, a KDE risk surface, and a LightGBM ensemble (70/30 blend, PR-AUC-gated), deployed on Azure and Vercel with automated CI/CD and predicted crime at an accuracy of 73%.
- Engineered a production ML pipeline with automated data validation (Great Expectations, 5 quality gates) and model validation (PR-AUC + recall@10% time-based holdout), logging all experiments to MLflow with a champion/challenger gate blocking underperforming models from deployment.
- Engineered end-to-end CI/CD pipelines using GitHub Actions for continuous integration (automated linting via Ruff, pytest test suite on every push) and continuous delivery to Azure Container Apps and Vercel; implemented MLflow-based machine learning model management with a weekly automated retrain, evaluate, and promote workflow.

### RAG PDF over Documents

- Built an end-to-end RAG pipeline with a modular 5-component architecture (PDF ingestion, chunking, Jina AI embedding, Qdrant vector search, and LLM generation), served via a FastAPI backend and a Streamlit frontend. Used FAISS for content embedding and retrieval.
- Implemented event-driven PDF ingestion using Inngest with idempotent vector upserts — re-ingesting the same document overwrites existing vectors, eliminating duplicate retrieval noise.

### Credit Card Dashboard

*MS Excel | Power BI*

- Analysed 656K credit card transactions across 4 card categories in Power BI, identifying that blue cardholders drove 83% of total revenue (46M/55M) and Swipe transactions accounted for 64% of volume.

- Built a dual-dashboard customer segmentation report revealing the 40–50 age group and high-income businessmen as top revenue drivers, with NY, TX, and CA leading across 5 states tracked.